

Rui Xie

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Affiliation: School of Computer Science, Shanghai Jiao Tong University

Program: First-year Direct Ph.D. Student in Computer Science; Advisor: Prof. Lu Chen



EDUCATION

Shanghai Jiao Tong University

2025.09 – Present

Direct-entry Ph.D. in Computer Science; Advisor: Prof. Lu Chen; GPA: 3.89/4.0

Shanghai Jiao Tong University

2021.09 – 2025.06

B.Eng. in Computer Science, School of Computer Science

Core-course average: 91.70/100 | Rank: 23/103 | Selected coursework: Discrete Mathematics (100), Compilers (100), Computer Architecture (99), Operating Systems (98), Data Structures (98), Probability and Statistics (99)

PUBLICATIONS

- [1] **ASIL: Replacing Screenshot-and-Click with Structured State and Semantic Actions**
[Rui Xie](#), Lu Chen*
[Findings of EMNLP 2026](#) | First Author
- [2] **GUIDE: Resolving Domain Bias in GUI Agents through Real-Time Web Video Retrieval and Plug-and-Play Annotation**
[Rui Xie](#), Zhi Gao, Chenrui Shi, Zirui Shang, Lu Chen*, Qing Li*
[ECCV 2026](#) | First Author
- [3] **MatToolBench: A Real-Environment Benchmark for Evaluating Multimodal Agents on Professional Materials Science Software**
Mei Wu[†], [Rui Xie](#)[†], Runyu Zhang, Lu Chen*, Bo Chen, Kai Yu*, Xin Chen* († equal contribution)
[Under Review](#) | Co-first Author
- [4] **TongUI: Internet-Scale Trajectories from Multimodal Web Tutorials for Generalized GUI Agents**
Bofei Zhang[†], Zirui Shang[†], Zhi Gao[†], Wang Zhang, [Rui Xie](#), Xiaojian Ma, Tao Yuan, Xinxiao Wu, Song-Chun Zhu, Qing Li*
[AAAI 2026](#) | Co-author
- [5] **Mobile-Env: Building Qualified Evaluation Benchmarks for LLM-GUI Interaction**
Danyang Zhang, Zhennan Shen, [Rui Xie](#), Situo Zhang, Tianbao Xie, Zihan Zhao, Siyuan Chen, Lu Chen, Hongshen Xu, Ruisheng Cao, Kai Yu
[Chinese Journal of Computers, Under Review](#) | Third Author

RESEARCH EXPERIENCE

Qwen Foundation Model Team: GUI Agent Post-training

2026.04 – Present

Qwen Foundation Model Team, Alibaba

Research Intern

- Participating in **post-training for Qwen mainline foundation models**, with a focus on improving agent capabilities for **computer use**.

ASIL: Structured States and Semantic Actions for Agent-Software Interaction

2026.05 – Present

X-LANCE Lab, Shanghai Jiao Tong University

First Author

- Proposed **ASIL**, replacing screenshot-and-click control with **structured software states** and **semantic actions**; built a shared-state benchmark spanning 15 applications, 300 single-application tasks, and 80 multi-application tasks.
- Implemented a unified **JSON observation-action protocol** across six realization patterns and a semi-automated onboarding framework for interface qualification, adapter generation, probing, and audit.
- Closed models scored above 80 with fewer than five actions per task; GPT-5.4 reached **81.6** within 15 actions versus **6.6** for screenshot GUI control within 50 steps, while ASIL SFT and RL raised Qwen3.5-9B from **66.6 to 82.2**.

RESEARCH EXPERIENCE (CONTINUED)

GUIDE: Video-Retrieval-Augmented Mitigation of Domain Bias in GUI Agents

2025.03 – Present

X-LANCE Lab, Shanghai Jiao Tong University / BIGAI

First Author

- Proposed **GUIDE**, a training-free, plug-and-play framework that addresses **planning** and **grounding** biases in GUI agents operating domain-specific software.
- Developed a subtitle-based **Video-RAG retrieval pipeline** with three-stage progressive retrieval (domain classification, topic extraction, and relevance matching). Built a **fully automated pairwise VLM video-annotation pipeline** that infers inter-frame actions from consecutive keyframes with enhanced UI-element detection, producing transferable planning and grounding knowledge.
- Validated on OSWorld: integration with multi-agent systems and diverse single-model agents **improved absolute success rates by 4.47%–7.48%** and reduced execution steps, without changing model parameters or architectures.

MatToolBench: Real-Environment Benchmarking for Multimodal Agents in Materials Science

2025.10 – Present

X-LANCE Lab, Shanghai Jiao Tong University / Suzhou Laboratory

Co-first Author

- Led the development of the first **real-environment GUI-agent benchmark** for professional materials science software, covering 10 tools and 204 tasks across GUI operation, code execution, and cross-tool interaction.
- Designed and implemented a **Windows-VM-based isolated test environment** with Docker deployment, parallel evaluation on Azure ML, standardized environment resets, and Gym interfaces. Introduced a **multi-level evaluation framework** spanning text recognition, visual-text matching, LLM-based aesthetic assessment, and code matching.
- Benchmarking state-of-the-art multimodal agents revealed substantial performance gaps for general-purpose agents in professional domains.

TongUI: Large-Scale GUI-Agent Data from Multimodal Web Tutorials

2024.10 – 2025.04

BIGAI / Shanghai Jiao Tong University

Co-author

- Contributed to **GUI-Net-1M**, an open-source dataset of 1 million GUI trajectories spanning 5 operating systems and more than 280 applications.
- Conducted ScreenSpot offline and MiniWeb online evaluations; contributed to Qwen2.5-VL **SFT** and the development of TongUI Agent, which outperformed baselines by more than 10% across multiple benchmarks.

BIGAI GUI Agent Internship: macOS Multi-App Systems and Reinforcement Learning

2025.03 – 2025.08

Beijing Institute for General Artificial Intelligence (BIGAI)

Research Intern

- Contributed to the development, testing, and deployment of a multi-application GUI agent for macOS, supporting task automation and cross-application collaboration across 10 desktop applications.
- Developed agents for **Browser, Calendar, and WeChat**; contributed to reinforcement-learning task generation and DAPO training integration for GUI agents.

Mobile-Env: Benchmarking LLM-GUI Interaction

2024.04 – 2024.11

X-LANCE Lab, Shanghai Jiao Tong University

Third Author

- Built an Android-based evaluation benchmark for LLM GUI agents; designed tasks for the Ele.me app and implemented traffic capture, local replay, and isolated environment deployment.
- Benchmarked mainstream language models and introduced chain-of-thought prompting to standardize agent behavior and improve icon-clicking accuracy.

TECHNICAL EXPERTISE

- Research:** LLM/VLM agents, with sustained experience in **GUI agents** spanning agent architectures, large-scale trajectory collection, model fine-tuning, reinforcement learning, benchmark development, and evaluation.
- Methods and Systems:** VLM SFT/LoRA fine-tuning and **reinforcement learning** (PPO, GRPO, and DAPO); cross-platform GUI-agent development and evaluation on Android, Windows, macOS, and Web.
- English:** CET-6, 564.

HONORS & AWARDS

Class of 1988 Alumni Care Fund Scholarship (4 recipients)

2024

Department of Computer Science, Shanghai Jiao Tong University